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Characteristics, behaviors and association between Human African Trypanosomiasis and HIV seropositivity among volunteer blood donors in a semi-rural area: A survey from Kikwit, the Democratic Republic of Congo

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Abstract

Blood safety is a major element in the strategy to control the HIV epidemic. The aim of this study was to determine the prevalence and the associated factors of a positive HIV test among blood donors and its association between Human African Trypanosomiasis in Kikwit, the Democratic Republic of Congo. A cross-sectional study was conducted between November 2012 and May 2013. An anonymous questionnaire was designed to extract relevant data. The average mean age of participants was 30 years. The majority were man (67.8%). The overall prevalence of HIV, syphilis, hepatitis B, hepatitis C and human African trypanosomiasis was respectively 3.2%, 1.9%, 1.6%, 1.3% and 1.3%. Alcohol intake, casual unprotected sex, not using condoms during casual sex, sex after alcohol intake and seroprevalence of human African trypanosomiasis were significantly associated with a positive HIV test result ($p<0.05$). In this study, sexual risk behaviors were the major risk factors associated with positive HIV tests in blood donors living in Kikwit. It is important to raise awareness about HIV and voluntary blood donation in response to some observations noted in this study such as the low educational level of the blood donors, the low level of knowledge of HIV prevention methods.

Keywords

Blood donors, HIV, Human African Trypanosomiasis behaviors, risk factors, rural area, Kikwit, the Democratic Republic of Congo, Africa

Introduction

Blood transfusion security is one of the strategies for disease control by the World Health Organization (WHO Expert Group, 2012). In the Democratic Republic of Congo (DRC), blood transfusion has been the focus of control efforts by the implementation of a national center for blood transfusion (Democratic Republic of Congo Ministry of Health/ Programme National de Transfusion Sanguine, 2006).

In different regions of the DRC, it has been documented that transfusion is often performed without screening of donor blood. The viral and parasitic contamination by this way has been reported in previous studies and the prevalence in blood donors varies from 1.4% to 4.7% (Batina *et al.* 2010; Tshilolo *et al.* 2007; Kabinda *et al.* 2014; Namululi 2013; Nzaji and Ilunga 2013; Batina 2007; Mbendi Nlombi *et al.* 2001).

However, the prevalence of HIV among blood donors in the province of Bandundu is unknown. The aim of this study was to determine the prevalence and the associated factors of a positive HIV test among blood donors and its association between Human African Trypanosomiasis in Kikwit, the large city of the Province.

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Materials and Methods

This was a cross-sectional study conducted between November 2012 and May 2013 in Kikwit, the DRC. Kikwit is a semi-rural city of province of Bandundu in the south-western part of the DRC.

The study population included voluntary blood donors above 18 years and less than 65 years of age. All participants were consecutively recruited. Four health facilities supported by the National Blood Transfusion Program (PNLS) were included: Hôpital général de référence de Kikwit Nord, Hôpital général de référence de Kikwit Sud, centre de santé de référence de Lunia, and Centre Provincial de Transfusion Sanguine (CPTS).

An anonymous questionnaire was administered by a trained medical team (3 physicians, 4 nurses and 2 laboratory technicians). The questionnaire was prepared in French and translated in Kikongo (the predominant language in Kikwit).

The structured questionnaire contained questions addressing: (i) demographic data (age, gender, marital status, education level, occupation, religion, number of trips in last 12 months, staying in a diamond mining area and / or Kinshasa in the past, alcohol intake; (ii) donation of blood (membership in an association of volunteer donors), sexually transmitted infection in last 12 months and sexual behavior.

Hemoglobin (Hb) level, Hepatite C Virus (HCV), Hepatite B virus (HBV), syphilis and human African trypanosomiasis (HAT) and weight were noted by the medical team.

Serology of HIV infection was made by using two antibody detection tests (DetermineR and UnigoldR); HBV by using the Quick™ tests, HbS Ag serum/Plasma/Whole Blood Strip; Hepatitis C infection by using the QuickTM tests, HCV

Table I. HIV prevalence by socio-demographic characteristics of volunteers and voluntary blood donors in Kikwit

Variables	n	% HIV	OR (95% CI)	P–value
Age (years)				0.55*
18–25	126	4	1.4 (0.4–4.5)	
26–65	246	2.9	1	
Gender				1.00*
Men	253	3.2	1	
Women	120	3.3	1.1 (0.3–3.4)$	
Geographical location				0.35*
North Kikwit	254	3.9	2.4 (0.5–11.1)	
South Kikwit	119	1.7	1	
Marital status				0.15
Married	167	1.8	1	
Others	206	4.4	2.5 (0.7–9.4)	
Education level				1.00*
Primary and secondary	292	3.4	1.4 (0.4–6.4)	
Tertiary	80	2.5	1	
Job category				0.06
unemployed	149	5.4	3.1 (1.0 –9.9)$	
Employee	223	1.8	1	
Having stayed in a diamond area and/or Kinshasa				0.52
No	153	3.9	1.5 (0.5–4.4)	
Yes	220	2.7	1	
Number of trips the last 12 months				1.00*
0	234	3.4	1.2 (0.4–3.8)$	
≥	1	139	2.9	
Religion				0.76
Catholic	201	3.5	1.2 (0.4–3.6)$	
Others	171	2.9	1	
Alcohol intake				0.02
No	184	1.1	1	
Yes	189	5.3	5.1 (1.1–23.5)	

*Fisher exact test$: Exact 95% CI

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Table II. HIV seroprevalence based on blood donations, sexually transmitted infections, HIV knowledge and sexual risk behaviours among volunteer blood donors in Kikwit

Variables	n	% HIV	OR (95% CI)	P value
Number of blood donations				0.55*
1	145	4.1	1.6 (0,5–4.8)[$]	
≥2	228	2.6	1	
Sampling location				0.21*
mobile collection	308	3.6	2.4 (0.3–18.7)	
fixed collection	65	1.5	1	
membership voluntary blood donors association				0.34*
No	108	4.6	1.8 (0.6–5.5)[$]	
Yes	265	2.6	1	
Casual sex in the last 12 months and condom use				0.02*
Casual unprotected sex	97	7.2	4.2 (1.4–12.8)[$]	
Others	274	1.8	1	
Sex after alcohol intake				0.02*
No	299	2.0	1	
yes	72	8.3	4.4 (1.5–13.5) [$]	
Knowledge of a selling point of condom				0.19*
No	105	1.0	1	
Yes	267	4.12	4.7 (0.6–35.1)	

*fisher exact test [$]: Exact 95% CI

Ag serum/Plasma/Whole Blood Strip; syphilis infection by using the QuickTM tests, Syphilis serum/Plasma/Whole Blood Strip; HAT was made by using the Card Agglutination Trypanosoma Test (CATT).

The study was assessed and approved by the Provincial Health Officer (the highest ranked medical authority in the Oriental Province). A written informed consent signed by participants was obtained before their inclusion.

The data were analyzed using STATA version 12.1. Descriptive analysis were used to obtain mean for quantitative and proportion for all the qualitative variables. Association between variables and HIV test was evaluated using Chi^2 tests and Fischer's exact tests. The comparison of continuous variables was calculated using the Student T test. Logistic regression analysis was used to identify independent variables associated with the risk factors of positive HIV test. Statistical significance level was set at $p < 0.05$.

Results

A total of 372 blood donors were recruited. The average age of participants was 30 years (range, 23–42 years). The majority of participants were males (67.8%).

Table III. Relationship between casual sex the last 12 months & use condoms and HIV prevalence by marital status among volunteer donors

Variables	n	% HIV	OR	95% CI
Married				
Casual sex in the last 12 months and condom use				
Casual unprotected sex	21	9.6	15.2	1.3–175.2
Others	145	0.8	1	
Others				
Casual sex in the last 12 months and condom use				
Casual unprotected sex	76	6.6	2.2	0.6–8.5
Others	129	3.1	1	

homogeneity test P-value : 0.17
Crude OR : 4.18 [1.3–13.5]
M-H OR : 3,19 [1.0–10.3] MH P value : 0.03

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The majority (59%) of participants had stayed in a diamond mining area and / or Kinshasa in the past. However, the majority (62.7%) had not moved from Kikwit in the last 12 months. The majority (54%) of the subjects were non Catholic-Christians. Almost half of the participants admitted to take regular alcoholic beverages.

A minority (26.2%) had had casual unprotected sex in the last 12 months and 45.5% did not use condoms.

A majority of blood donors (61%) had at least two blood donations.

The mean weight was 61.3 kg (SD=10.6) and the mean Hb concentration was 14.0 g/l (SD= 1.2).

Among the blood donors, the overall prevalence rate of HIV was 3.2% (95% CI: 0.02–0.06). The prevalence rate of syphilis, hepatitis B, hepatitis C and human African trypanosomiasis was respectively 1.9%, 1.6%, 1.3% and 1.3%.

Alcohol intake, casual unprotected sex in the last 12 months, not using condoms during casual sex, sex after alcohol intake and seroprevalence of human African trypanosomiasis were significantly associated with positive HIV test result (Table I and II).

In this cohort, marital status was a confounding effect for relationship for sex in the last 12 months and condom use and of HIV seroprevalence (Table III).

Discussion

In our cohort, the median age was similar in comparison with others African series where the mean age varies from 28 to 29 years (Tagny *et al.* 2010).

In this cohort, the majority of donors were men. Males were also predominant in previous studies in Africa where the frequency varies from 68.3% to 80.1% (Malaki *et al.* 2013; Mole *et al.* 2011).

The majority of volunteer donors had made more than one donation and were member of an association of volunteer donors. This observation was comparable to that of African countries where the frequency varies to 55% to 67% (Tagny *et al.* 2009).

In this cohort, 3.2% were positive for HIV testing. Previous studies in the DRC showed that this prevalence varies from 1 to 2.9% in blood donors (Namululi *et al.* 2012; Namululi *et al.* 2013; Nzaji and Ilunga 2013; Batina *et al.* 2007). In comparison with neighboring countries, our prevalence was lower than the prevalence of 4.1% reported in Uganda, 5.1% reported in Rwanda and 13.5% in Central African Republic (Tagny *et al.*, 2010).

HBV antigens were detected in 1.6% of blood donors. This prevalence was lower than the 3% found in the east of the DRC where the prevalence varies from 3% to 4.2% (Batina *et al.* 2007; Kabinda *et al.* 2014; Namululi *et al.* 2012; Namululi *et al.* 2013). At the same time, our prevalence was similar to the prevalence of 1.6% reported in the South East of the DRC (Nzaji and Ilunga 2013).

The frequency of HCV antigens in this study (1.3%) was lower than that reported by Kabinda in the Eastern DRC with 3.8% of affected blood donors , but higher than that reported by Nzaji in the South East part of the DRC with 0.2% of affected blood donors (Nzaji and Ilunga, 2013).

The highest frequencies (60%) of west HAT in the world occur in the DRC (Lutumba *et al.* 2005). The province of Bandundu have the highest levels of HAT endemicity in the DRC (Hasker *et al.* 2012). No prevalence rate in blood donors could be found with the use of available computer-assisted medical literature search programs.

In the present study, the overall prevalence of syphilis was 1.9%. The prevalence varies from 0.2% to 1.1% in other part of the DRC (Nzaji and Ilunga 2013; Batina *et al.* 2007).

The results of this study show a significant association between HAT and the occurrence of positive HIV test. This observation has not been confirmed by subsequent studies. In Ivory Coast, it was found that there was no significant difference between HAT and the positive HIV test (Meda *et al.* 1995). HAT decreases the specificity of antibody detection tests for the diagnosis of HIV, which is a risk of false positive response to diagnostic tests, including HIV tests (Lejon *et al.* 2010).

Sex after alcohol intake, alcohol intake and the occasional sex in the last 12 months and condom use were identified to be related to risky sexual behavior. Our findings confirm and extend the findings of several studies (Mole *et al.* 2011; McFarland *et al.* 1998; Minga *et al.* 2005; Van Tieu and Koblin 2009).

Marital status was a confounding factor in relation with to sexual relationship the last 12 months, condom use and HIV seroprevalence. In Burundi and Zimbabwe, marital status was a factor significantly associated with HIV sepositivity in blood donors (McFarland *et al.* 1998; ONUSIDA, OMS 2009).

In our report, there were no significant association between gender and the positive HIV test. This observation has not been confirmed by subsequent studies carried out in 9 African countries (Gouws *et al.* 2008).

In this study, sexual risk behaviors were the major risk factors associated with positive HIV tests in blood donors. It is important to raise awareness about HIV and voluntary blood donation in response to some observations noted in this study such as the low educational level of the blood donors, the low level of knowledge for HIV prevention methods.

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